

Section 1: Given $f(x) = -3x^3$ and $g(x) = x^2 - 5$ find the following.

1. $f(g(2))$

2. $[g \circ f](2)$

Section 2: Find $[f \circ g](x)$ and $[g \circ f](x)$.

3. $f(x) = \frac{1}{2}x - 4$

4. $f(x) = 5x$

$$g(x) = 4x + 24$$

$$g(x) = 2x^3 + 3x^2 + x - 1$$

Section 3: Use $f(g(x))$ and $g(f(x))$ to determine if f and g are inverses. Show all work!

5. $f(x) = 3x^2 - 11$

$$g(x) = \sqrt{\frac{x+11}{3}}$$

Section 4: Find $f^{-1}(x)$ for the following functions

6. $f(x) = x^2 + 7$

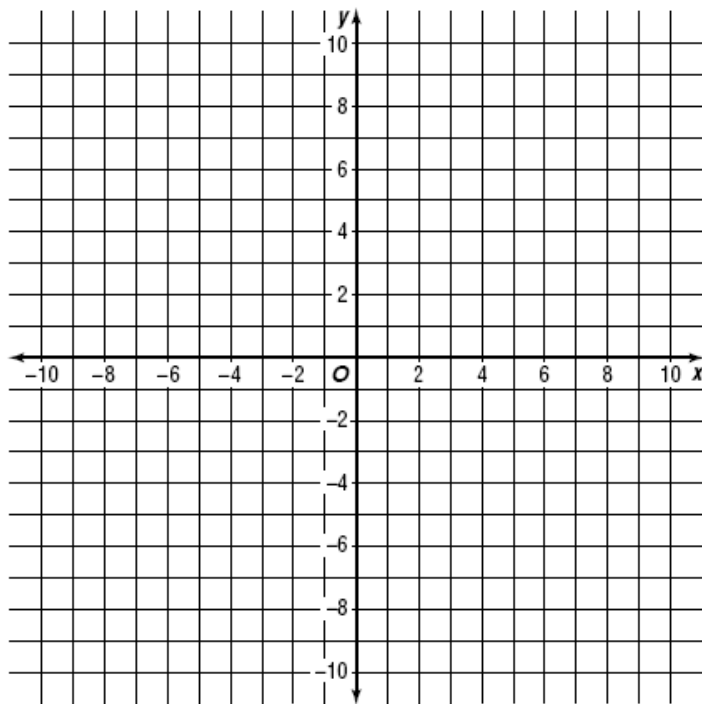
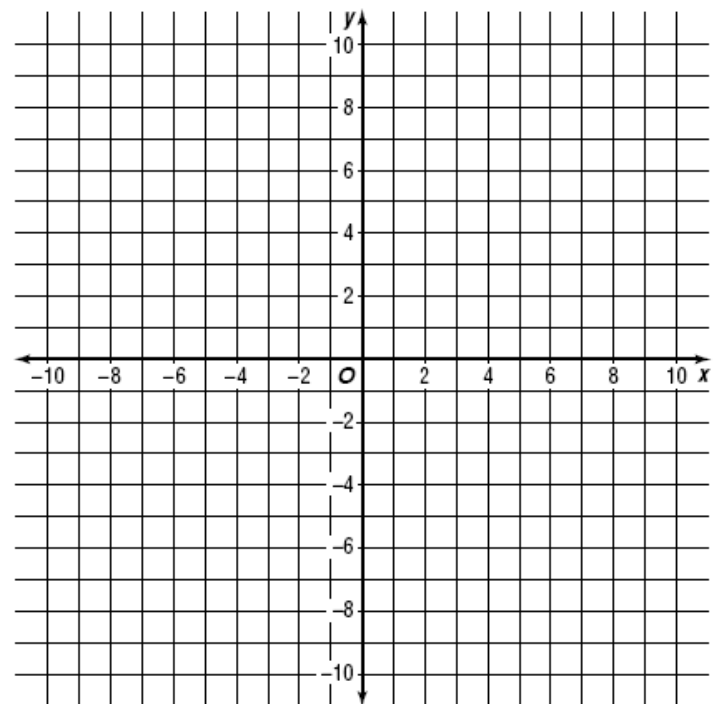
7. $f(x) = \sqrt[3]{\frac{x-2}{5}}$

1. The slope of a _____ line is zero.

2. Two lines are _____ if they intersect to form right angles.

Graph a line parallel to $y = -2x + 1$ that passes through $(-4, 0)$

Graph a line perpendicular to $y = -2x + 1$ that passes through $(0, 0)$



Express in scientific notation

0.0000456

78,000,000

Simplify:

$$\frac{1.4 \times 10^4}{0.2 \times 10^2}$$

$$\frac{3.3 \times 10^{-3}}{1.1 \times 10^{-5}}$$

$$(2 \times 10^4)(4 \times 10^{-5})$$

$$(7 \times 10^3)(3 \times 10^{-1})$$

1) $\log_2 16 - \log_3 81$

2) $\log_5 125 + \log_4 16 - \log_2 32$

Solve

1. $\log_9 \frac{1}{9} = x$

2. $\log_3 3 + \log_2 8 = x$

3. $\log_9 (6x + 7) = 2$

4. $\log_6 36 = 5x + 3$

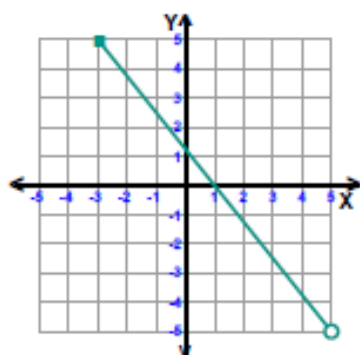
5. $\log x = -2$

6. $\log_5 \sqrt{5} = x$

7. $\log_6 \frac{1}{36} =$

8. $\log_8 4 =$

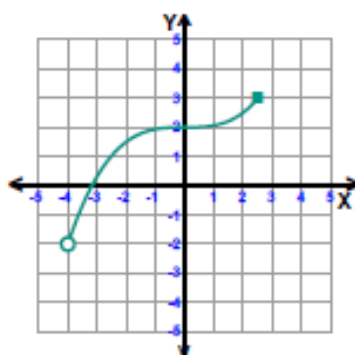
1)



Domain: _____

Range: _____

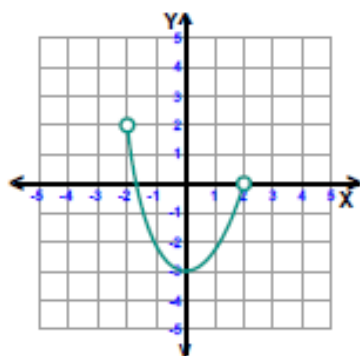
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Domain: _____

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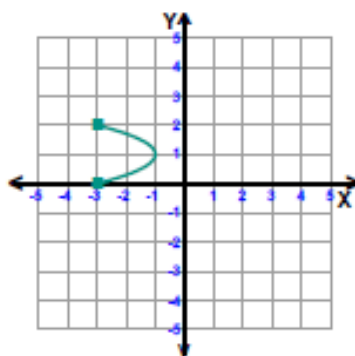
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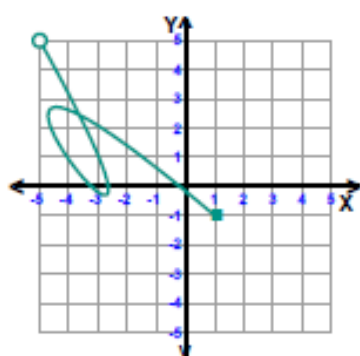
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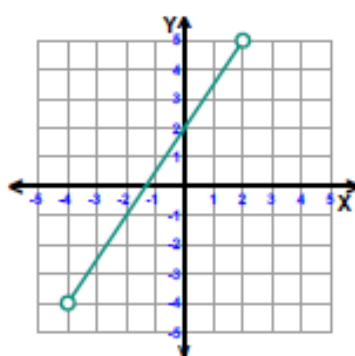
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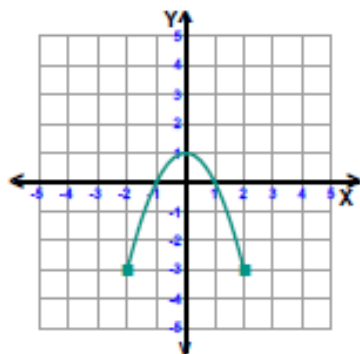
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Domain: _____

Range: _____

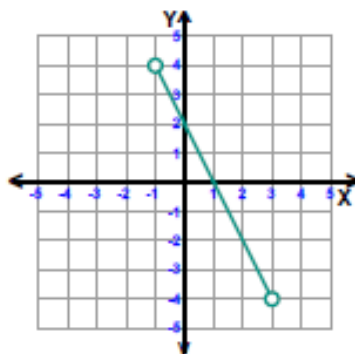
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Domain: _____

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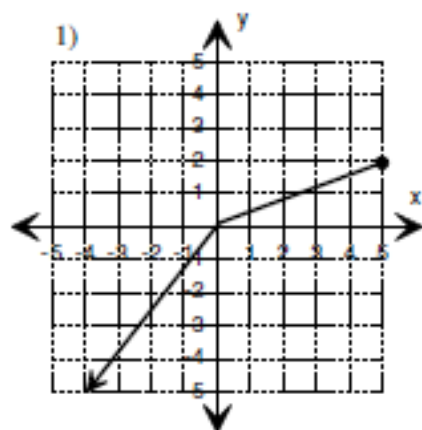
8)



Domain: _____

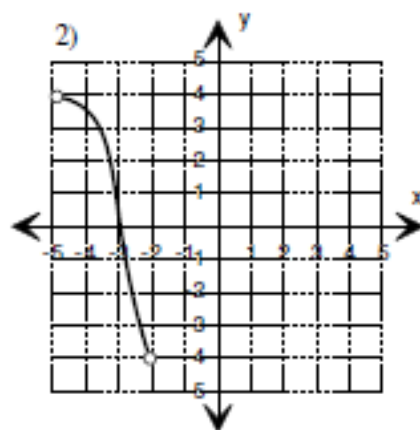
Range: _____

Find the Domain and Range for each graph.



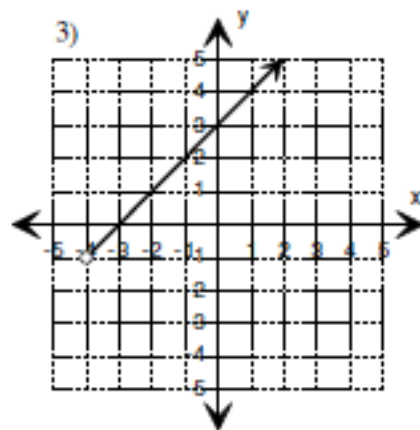
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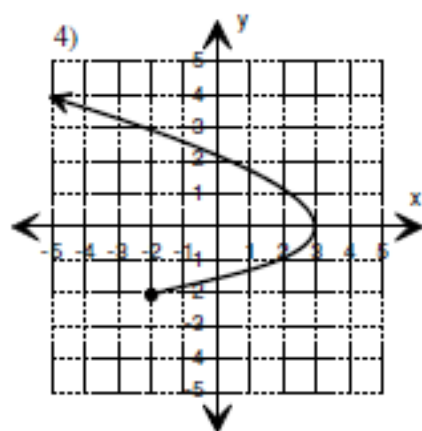
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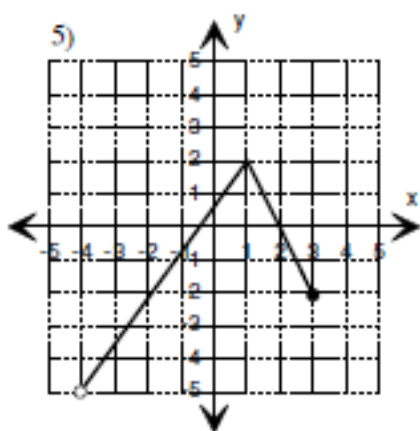
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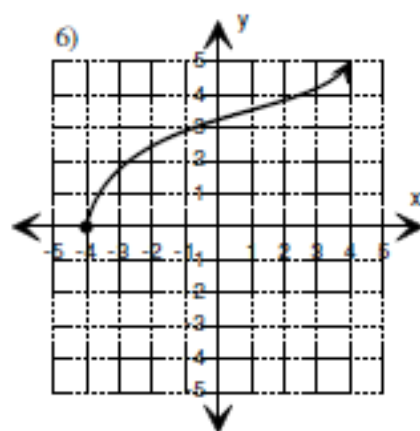
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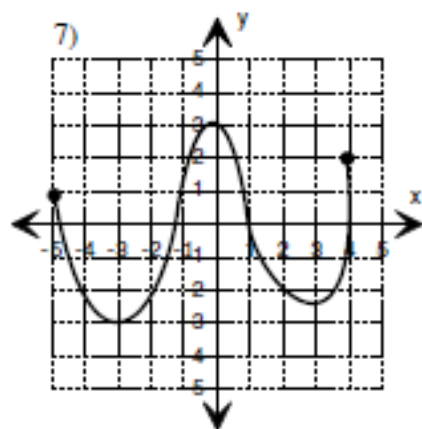
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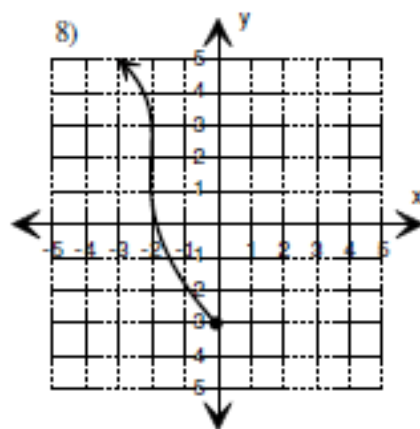
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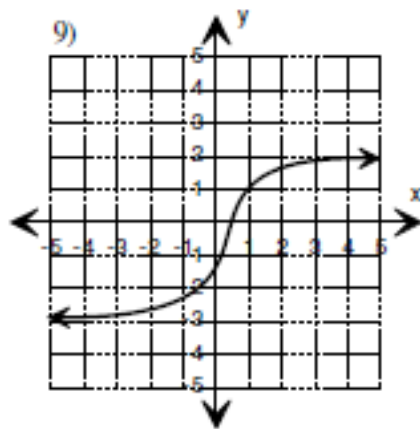
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Domain : _____

Range : _____



Domain : _____

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